THE NAIVE COTANGENT COMPLEX OF A FINITELY PRESENTED MORPHISM IS ETALE LOCALLY A TWO-TERM COMPLEX OF FREE MODULES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5425-5438 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 5440-5448, the complete author proof (7 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The naive cotangent complex of a morphism, chart presentations of algebraic spaces, and etale localisation compatibility; the reduction is retained.

Source reference. Original lemma lemma-NL-compare-spaces-schemes: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5324–5334. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-NL-etale-localization: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5297–5308. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-lemma-netherlander-fp: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5448–5448. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is locally of finite presentation, then $NL_{X/Y}$ is étale locally on X quasi-isomorphic to a complex*

$$\ldots \to 0 \to \mathcal{F}^{-1} \to \mathcal{F}^0 \to 0 \to \ldots$$

of quasi-coherent $\mathcal{O}_X$-modules with $\mathcal{F}^0$ of finite presentation and $\mathcal{F}^{-1}$ of finite type.

Proof. Formation of the naive cotangent complex commutes with étale localization by Lemma lemma-NL-etale-localization. This reduces us to the case of schemes by Lemma lemma-NL-compare-spaces-schemes. The result in the case of schemes is More on Morphisms, Lemma more-morphisms-lemma-netherlander-fp. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.